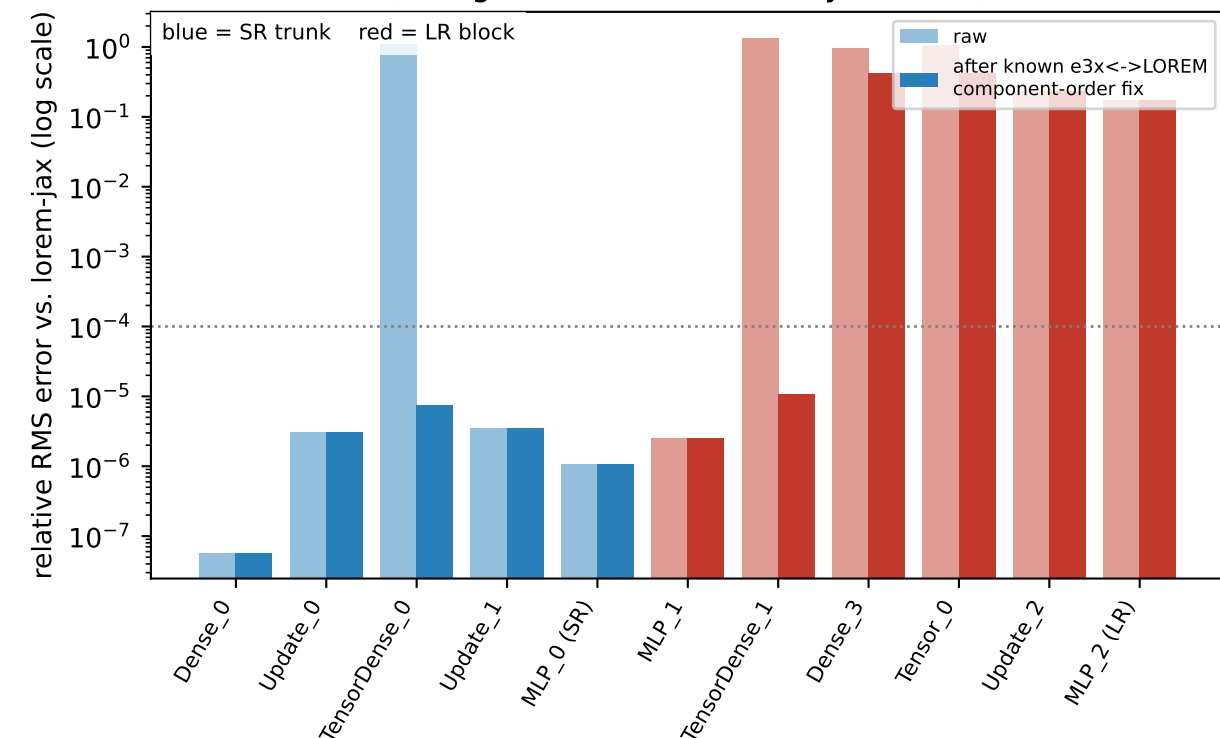
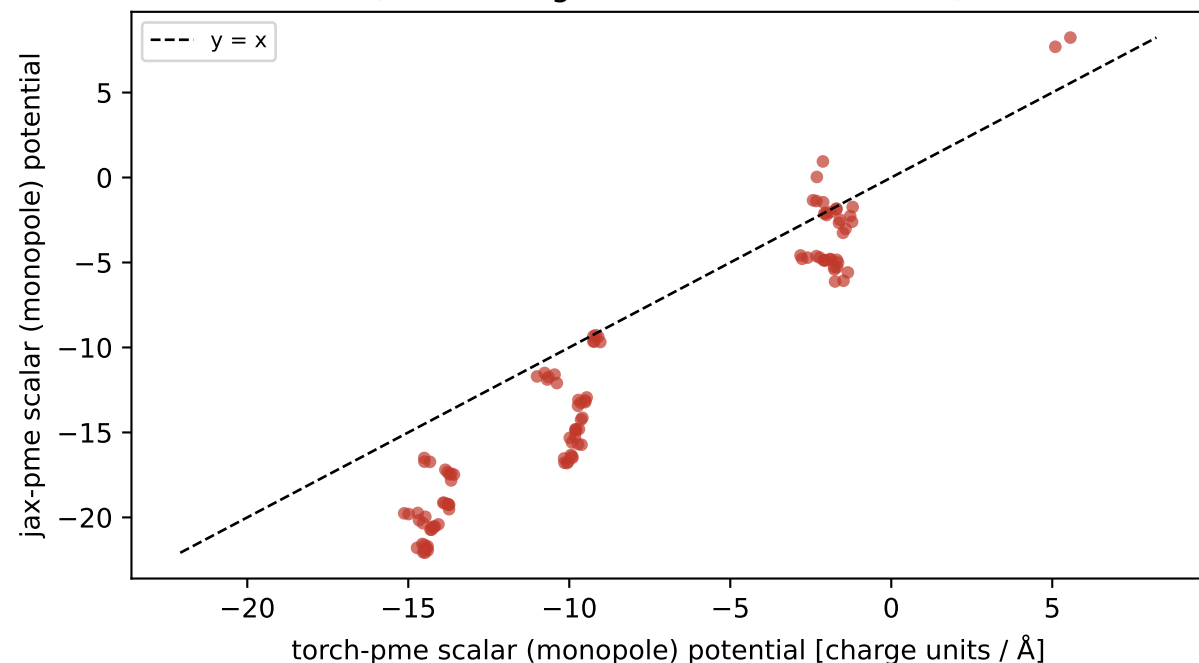


LOREM torch port vs. lorem-jax checkpoint (AuMgO): where the ~ 0.7 meV/atom gap comes from

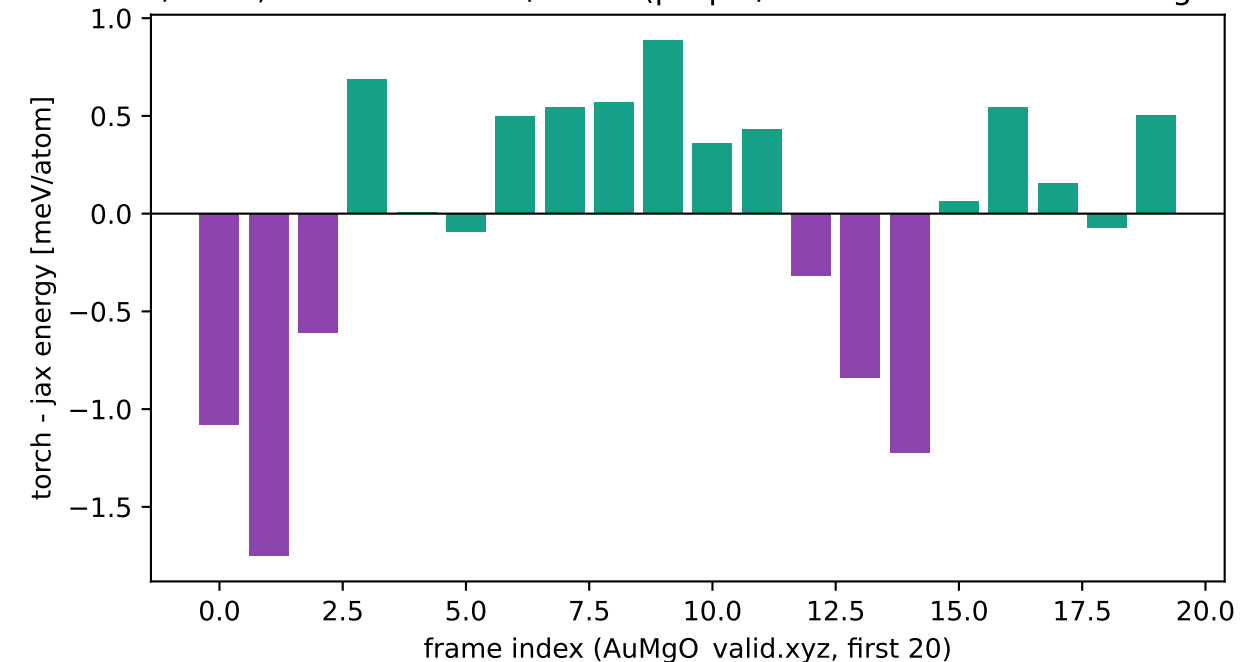
Where the torch reimplement diverges from lorem-jax
(AuMgO frame 0, module-by-module)



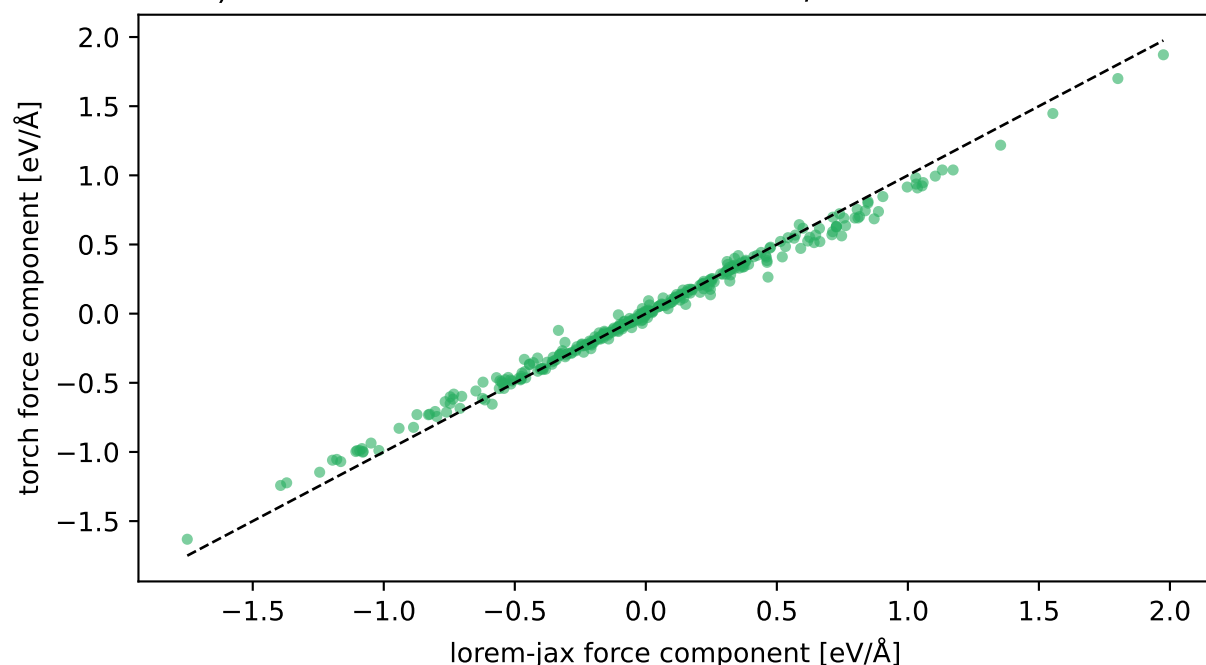
Ewald potential itself disagrees
(same charges in, $r = 0.965$, not 1:1)



Per-frame energy residual
-0.71 meV/atom, MAE=0.56 meV/atom (purple/teal = the two structural groups in this slice)



Force components, frame 0
RMSE = 60.0 meV/Å



Conclusion: the equivariant SR trunk and LR charge heads are numerically exact ($\sim 1e-6$ relative, floating-point-level) once the known, benign e3x <-> LOREM spherical-component ordering is accounted for. The real, unresolved gap is the Ewald long-range potential itself (torch-pme vs. jax-pme) -- it disagrees even on the simple scalar/monopole channel with identical input charges.